FORMAL FUNCTIONS FOR A PRINCIPAL IDEAL: DERIVED COMPLETION OF RΓ

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Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- RΓ commutes with R lim. Cohomology of Sheaves, Lemma cohomology-lemma-Rf-commutes-with-Rlim. RΓ(X, R lim E_n) = R lim RΓ(X,E_n).
- Completion vs derived completion. More on Algebra, Lemma more-algebra-lemma-compare-completion-derived-completion. The commutative diagram with the f-torsion T_f.

Source reference. Original reference 0BKG: sources/stacks-chow/tags.txt, source lines 14559–14559. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0BKP: sources/stacks-chow/tags.txt, source lines 14567–14567. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0H32: sources/stacks-chow/tags.txt, source lines 21302–21302. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Let $A \to \Gamma(X, \mathcal{O}_X)$ be a ring map and let $f \in A$. Let E be an object of $D(\mathcal{O}_X)$. Denote*

$$E_n = E \otimes_{\mathcal{O}_X} (\mathcal{O}_X \xrightarrow{f^n} \mathcal{O}_X)$$

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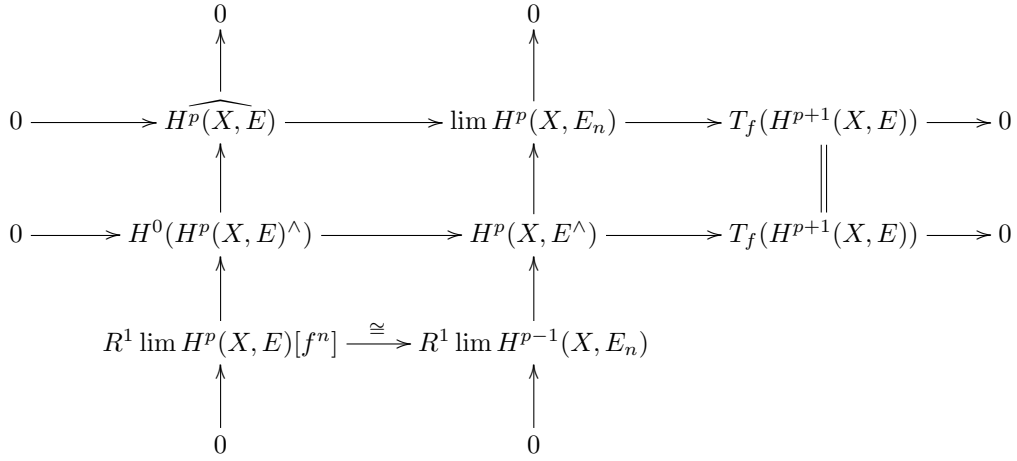
and set $E^\wedge = R\lim E_n$. *For* $p \in \mathbf{Z}$ *is a canonical commutative diagram*



with exact rows and columns where $\widehat{H^p(X,E)} = \lim H^p(X,E)/f^n H^p(X,E)$ *is the usual* f*-adic completion,* $H^p(X,E)^\wedge$ *is the derived* f*-adic completion, and* $T_f(H^{p+1}(X,E))$ *is the* f*-adic Tate module, see More on Algebra, Example 0BKG. Finally, we have* $H^p(X,E^\wedge) = H^p(R\Gamma(X,E)^\wedge)$.

Proof. Observe that $R\Gamma(X,E^\wedge) = R\lim R\Gamma(X,E_n)$ by Lemma 0BKP. On the other hand, we have

$$R\Gamma(X,E_n) = R\Gamma(X,E) \otimes^{\mathbf{L}}_A (A \xrightarrow{f^n} A)$$

(details omitted). We find that $R\Gamma(X,E^\wedge)$ is the derived f-adic completion $R\Gamma(X,E)^\wedge$. Whence the diagram by More on Algebra, Lemma 0H32. □